# QUESTION-TO-SLIDE MAPPING

## SECTION B - LECTURE 03

# Elements of Climate: Temperature, Humidity, Vapour Pressure & Precipitation

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| **Course** | BECM 3115 - Climate and Architectural Design |
| **Lecture deck** | 3.0 - Elements of Climate (10 PDF pages/slides) |
| **Question bank** | Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present) |
| **Mapping rule** | Keep Lecture 03 separate from Lecture 03.1. Map only the content actually developed here: climatic-information workflow, temperature recording/Stevenson Screen, humidity/RH, vapour pressure and precipitation. Driving rain, solar-radiation measurement, wind/wind rose, sky condition and vegetation are deferred to Lecture 03.1. |

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| **RESULT** | Lecture 03 is the measurement and basic climatic-data foundation. Its strongest repeated match is temperature measurement / Stevenson Screen (2017, 2018, 2021, 2022 and 2024). The deck also directly supports the “elements of climatic information” question and part of the humidity/RH and atmospheric-pressure families. |

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

# 1. Executive Summary

The lecture begins by listing the climatic elements relevant to building designers and explaining that raw meteorological records must be analysed for beneficial/harmful building effects. It then develops temperature measurement and recording, Stevenson Screen conditions, absolute/relative humidity and hygrometer use, vapour pressure, and precipitation/rainfall data. The lecture stops before the detailed driving-rain, solar, wind, sky-condition and vegetation treatment, which is why Lecture 03.1 must remain separate.

| 10 | 11 | 5 | 0 | 6 | 93 |
|---|---|---|---|---|---|
| Slides reviewed | Question links | Exact matches | Direct matches | Partial matches | Mapped marks |

## Highest-priority slide blocks

| Slides | Lecture block | Historical signals | Exam value |
|---|---|---|---|
| 4-5 | Temperature data + Stevenson Screen + thermograph | 2017, 2018, 2021, 2022, 2024 | Highest priority repeated measurement block. |
| 6-7 | Absolute/Relative Humidity + wet/dry-bulb hygrometer + data analysis | 2020 RH question; 2024 measurement | Useful direct support; numerical SH formula is not fully printed. |
| 2-3 | Elements of climate + raw data -> design analysis | 2021 Q2(d); 2024 Q4(b) | Strong conceptual block for climatic information. |
| 8 | Vapour pressure and P = Pa + Pv | 2022/2023 atmospheric-pressure questions partial | Useful relation, but total atmospheric pressure is not fully developed. |
| 9 | Precipitation, rain gauge, monthly/24 h/hourly data | Supports driving-rain/drainage topics | Good definition/data page; driving rain itself belongs to 03.1. |

| Priority note | Memorize Slides 4-6 first. For any field-measurement answer, write parameter -> instrument -> exposure/height condition -> data to record/analyse. Keep Slide 2 as the master climatic-element checklist. |
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## 2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

| Year | Original | Marks | QB p. | Question / task | Match | Lecture slides | Coverage assessment |
|---|---|---|---|---|---|---|---|
| 2017 | Q7(a)(iii) | 5* | 2 | Write short note on Stevenson Screen. | Exact | 4-5 | The deck gives true-air-temperature conditions, louvered box, shade and mounting height; Slide 5 shows the instrument visually. |
| 2018 | Q3(b) | 8 | 3 | Show the working mechanism of Stevenson Screen for measuring temperature. | Exact | 4-5 | Direct measurement/installation content; the student can explain how the screen shields radiation while allowing air exposure. |
| 2020 | Q1(b) | 15 | 5 | Describe how humidity level in air depends on site/environmental factors and show their reciprocal relationship. | Partial | 6-7 | The lecture defines and measures humidity but does not explain the requested soil, water-body, altitude, precipitation, airflow and vegetation interactions. |
| 2020 | Q2(b) | 12 | 5 | Define Absolute Humidity and Saturation Point Humidity; calculate RH for given values. | Partial | 6 | AH and RH are defined, but Saturation Humidity is not named separately and the numerical formula is not explicitly written on the slide. |
| 2021 | Q2(d) | 5 | 6 | Outline elements of climatic information. | Exact | 2-3 | Slide 2 lists the designer-relevant climatic elements and Slide 3 explains why raw climatic data must be analysed. |
| 2021 | Q3(c)(ii) | 5 | 6 | Write short note on Stevenson Screen. | Exact | 4-5 | Direct short-note match. |

## 2. Detailed Year-wise Question Mapping - continued

| Year | Original | Marks | QB p. | Question / task | Match | Lecture slides | Coverage assessment |
|---|---|---|---|---|---|---|---|
| 2022 | Q3(a) | 5 | 8 | Briefly describe atmospheric pressure. | Partial | 8 | Slide 8 gives vapour partial pressure and P = Pa + Pv, but does not provide a complete standalone atmospheric-pressure explanation. |
| 2022 | Q3(b) | 10 | 8 | Write down the instrument required to measure temperature data in an open field and discuss its working mechanism. | Exact | 4-5 | Thermometer/thermograph and Stevenson Screen placement are directly specified. |
| 2023 | Q2(d) | 5 | 10 | Briefly describe atmospheric pressure. | Partial | 8 | Same limitation as 2022: the relation is present but the full atmospheric-pressure concept is not developed. |
| 2024 | Q4(b) | 13 | 13 | What climatic factors affect building design and how do they influence it? | Partial | 2-3 | The factor list and need for analysis are present, but individual design effects require Lecture 03.1 and later design lectures. |
| 2024 | Q7(c) | 10 | 15 | Describe the process of measuring climatic parameters - temperature, humidity and air flow - at a specific site. | Partial | 4-7 | Temperature and humidity measurement are well covered; airflow measurement belongs to Lecture 03.1. |

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| **Coverage key** | Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required. |

## 3. Repeated Question Patterns

The lecture is most valuable when the exam asks “how do you measure/record climatic data?” rather than “how does climate behave globally?” The same measurement logic repeats: protect the sensor from bias, use the correct instrument, and analyse maxima/minima/means rather than quoting one reading.

| P | Normalized repeated family | Years seen | Slides | Why it matters |
|---|---|---|---|---|
| S | Temperature measurement / Stevenson Screen / field instrument | 2017, 2018, 2021, 2022, 2024 | 4-5 | Most repeated direct family from this deck. |
| A | Elements of climatic information / designer data analysis | 2021, 2024 | 2-3 | Good theory block and answer introduction. |
| A | Humidity / AH-RH / humidity measurement | 2020, 2024 (+2021 numerical elsewhere) | 6-7 | Useful but the full SH numerical chain needs another source. |
| B | Atmospheric vs vapour pressure | 2022, 2023 | 8 | Only partial support; do not overclaim. |
| B | Precipitation data and drainage significance | Support to DRI/drainage | 9 | Foundation for Lecture 03.1 driving-rain work. |

## 4. Quick Recognition Rules

| If the question says... | Go to slides | Recognition rule |
|---|---|---|
| “Stevenson Screen / true air temperature / open field instrument” | 4-5 | Shade + louvered box + about 1.20-1.80 m above ground; thermometer/thermograph inside. |
| “elements of climatic information” | 2-3 | Temperature, humidity/vapour pressure, precipitation/driving rain, solar, wind, sky condition, vegetation. |
| “absolute / relative humidity” | 6-7 | AH = actual moisture; RH = actual / capacity at same temperature x 100%; wet/dry bulb + psychrometric chart. |
| “vapour / atmospheric pressure” | 8 | P = Pa + Pv is present, but use the proper pressure lecture/note for a full atmospheric-pressure answer. |
| “precipitation / drainage rainfall data” | 9 | Rain gauge; mm per time; monthly wet/dry pattern; 24-h flood guide; hourly intensity for drainage. |
| “airflow measurement / driving rain / wind rose” | Lecture 03.1 | Do not force these into Lecture 03; they are taught in the next separate deck. |

# 5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

| Slides | What the slides teach | Past-question links | Study action |
|---|---|---|---|
| 2 | Designer-relevant elements of climate | 2021 Q2(d); 2024 Q4(b) partial | Memorize the 7-element list. |
| 3 | Raw meteorological data must be analysed for building benefit/harm | 2021 Q2(d); 2024 Q4(b) | Use as design-data workflow sentence. |
| 4-5 | Temperature measurement, thermograph, Stevenson Screen, data summaries | 2017 Q7(a)(iii); 2018 Q3(b); 2021 Q3(c)(ii); 2022 Q3(b); 2024 Q7(c) | Highest priority; practise a labelled Stevenson Screen sketch. |
| 6-7 | AH, RH, hygrometer, psychrometric use, humidity records | 2020 Q1(b), Q2(b); 2024 Q7(c) | Know definitions + instruments; flag SH numerical gap. |
| 8 | Vapour pressure relation | 2022 Q3(a); 2023 Q2(d) partial | Memorize $P = Pa + Pv$ but do not claim full atmospheric-pressure coverage. |
| 9 | Precipitation definition/measurement and design data | Supports DRI and drainage questions | Connect immediately to Lecture 03.1 Driving Rain. |
| 10 | Closing | None | Skip. |

# 6. Coverage Gaps & Diagnostic Exclusions

The slide title on Page 2 lists seven climatic elements, but this PDF only develops temperature, humidity/vapour pressure and precipitation. Treating the list itself as full coverage would produce incomplete exam answers.

| Gap | Affected questions | Action |
| --- | --- | --- |
| Driving rain / DRI is only named, not explained | 2021-2024 DRI family | Use Lecture 03.1 Slides 2-4. |
| Wind measurement / wind rose is not developed | 2017/2024 wind-frequency graph; 2024 Q7(c) airflow | Use Lecture 03.1 Slides 8-11. |
| Solar duration/intensity, sky condition and vegetation are not developed | 2021 solar quantity; 2024 climatic-factor question | Use Lecture 03.1 and Lecture 02 as appropriate. |
| Saturation Humidity and explicit RH numerical equation are not printed | 2020 Q2(b); 2021 RH numerical | Use the uploaded class note / Section A humidity material; label it as supporting source. |
| Atmospheric pressure is broader than vapour pressure | 2022 Q3(a); 2023 Q2(d) | Slide 8 gives only the partial-pressure relation; supplement from the correct source. |

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| **Mapping discipline** | Lecture 03.0 and 03.1 are deliberately kept separate. A topic appearing in the Slide 2 checklist is not treated as taught unless later slides actually develop it. This prevents false Exact matches. |

# 7. Recommended Study Plan

This deck is short. Finish it in one sitting, then immediately pair it with Lecture 03.1 so the full Elements-of-Climate set is covered without mixing the two reports.

| Step | Study block | Slides | Practice immediately |
|---|---|---|---|
| 1 | Stevenson Screen + temperature data | 4-5 | 2018 Q3(b), 2022 Q3(b), 2021 short note. |
| 2 | Elements + climatic-data workflow | 2-3 | 2021 Q2(d), 2024 Q4(b) opening. |
| 3 | Humidity measurement / RH concept | 6-7 | 2020 Q2(b) with supporting formula source; 2024 Q7(c). |
| 4 | Vapour pressure | 8 | Prepare as a short definition + P = Pa + Pv, but flag atmospheric-pressure gap. |
| 5 | Precipitation | 9 | Write definition, instrument, units, flood/drainage significance; then move to 03.1 DRI. |

## Exam-ready minimum

| Priority | What to reproduce | Exam technique |
|---|---|---|
| A | Stevenson Screen sketch | Label louvers, shaded white box, thermometer and 1.2-1.8 m mounting height. |
| A | Climatic element list | Write all seven before discussing any specific element. |
| A | Humidity instrument line | Wet/dry-bulb hygrometer + psychrometric chart; continuous record by hygrograph. |
| B | Precipitation design data | Monthly pattern -> wet/dry season; 24-h max -> flood; hourly max -> drainage. |

**Bottom line** Lecture 03 is a measurement-heavy scoring deck, especially for Stevenson Screen. Its main trap is assuming that every element named on Slide 2 is fully covered here; wind, driving rain, solar, sky and vegetation belong to the separate 03.1 report.